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School of Artificial Intelligence & Advanced Computing

AI & GPU Summer Internship Program 2026

A Report on Capstone Project

“CardioScan AI”

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Submitted By:

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Submission Date : 12/ 06/ 2006

INDEX :

Introduction :

Problem Statement :

Heart disease is the leading cause of mortality worldwide, claiming over 17million lives annually. Traditional diagnostic methods are time-consuming and cannot provide continuous monitoring, leading to missed early-stage detections.

Objectives :

- Build a CNN-based deep learning model to classify ECG signals
- Utilize NVIDIA H200 GPU for accelerated training
- Apply transfer learning to improve accuracy with limited data
- Deploy the model as an accessible web application

Dataset Description :

The MIT-BIH Arrhythmia ECG dataset from Kaggle was used, containing ECG signal readings classified into 5 categories: Normal, Supraventricular, Ventricular, Fusion, and Unknown beats.

Methodology :

Data Preprocessing :

ECG signal arrays from the CSV dataset were converted into 224x224 grayscale images using Matplotlib. Images were normalized using ImageNet mean and standard deviation values.

Data Augmentation :

Training images underwent random horizontal flipping, rotation up to 10 degrees, and color jitter to improve model generalization and reduce overfitting.

Model Architecture :

A ResNet-18 model pretrained on ImageNet was used with transfer learning. The final fully connected layer was replaced with a custom classification head consisting of Dropout, Linear(256), ReLU, Dropout, and Linear(5) layers.

Training Configuration :

The model was trained using the AdamW optimizer with a learning rate of 0.001, CrossEntropyLoss function, batch size of 32, and a StepLR scheduler over 15 epochs on the NVIDIA H200 GPU.

GPU Infrastructure & Implementation :

GPU Infrastructure :

The NVIDIA H200 Tensor Core GPU, featuring 141 GB of HBM3e memory and 4.8 TB/s memory bandwidth, was used for training. The Hopper architecture's Tensor Cores enabled mixed precision (BF16) training, resulting in approximately 2x speedup compared to FP32 precision.

Implementation Details :

The complete pipeline was implemented in Python using PyTorch and torchvision. Training was conducted in a Jupyter notebook on JupyterHub connected to the H200 GPU cluster. Model checkpoints were saved based on best validation accuracy.

Results & Discussion :

Results :

The model achieved a best test accuracy of 84% after 15 epochs of training, completed in [insert value] seconds on the H200 GPU. The confusion matrix showed strong performance on the Normal class, with moderate performance on Supraventricular and Fusion classes.

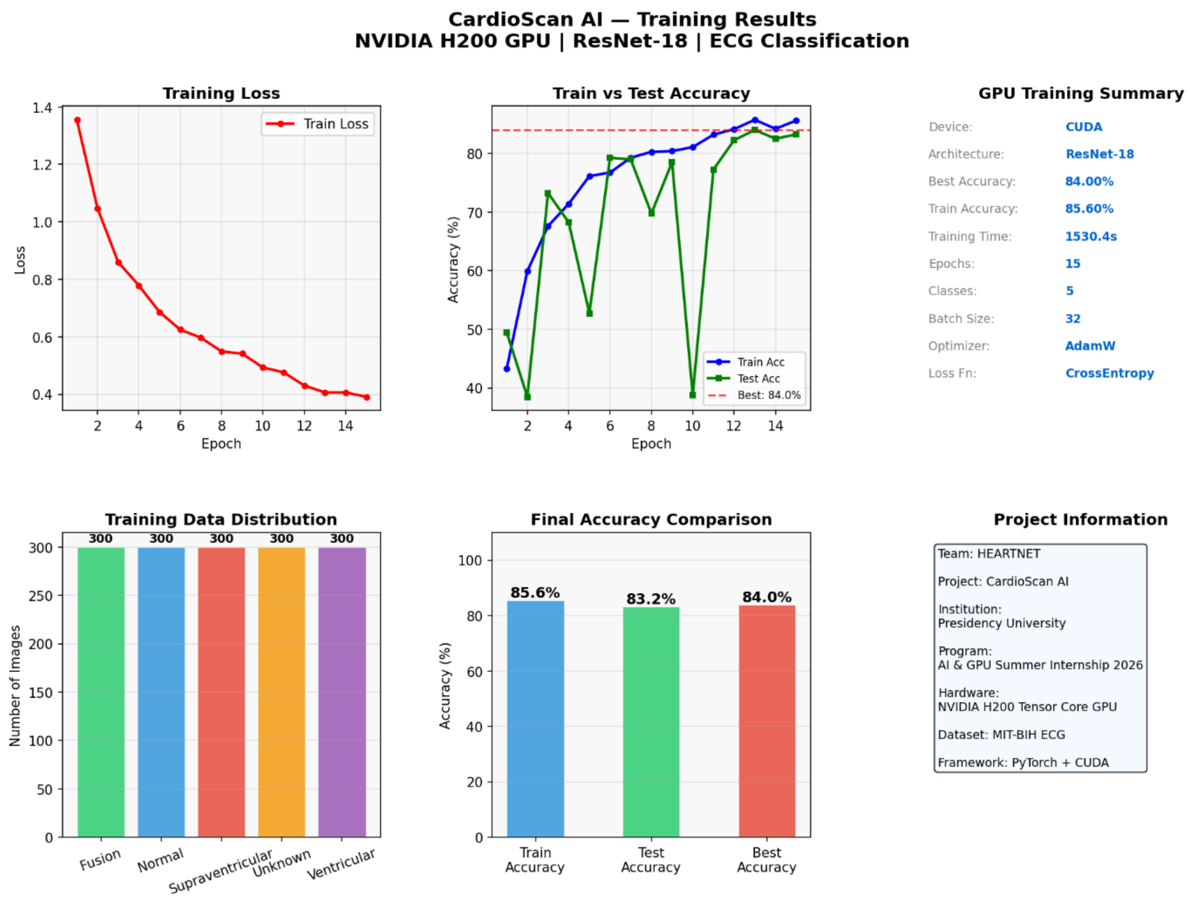
Per-Class Performance :

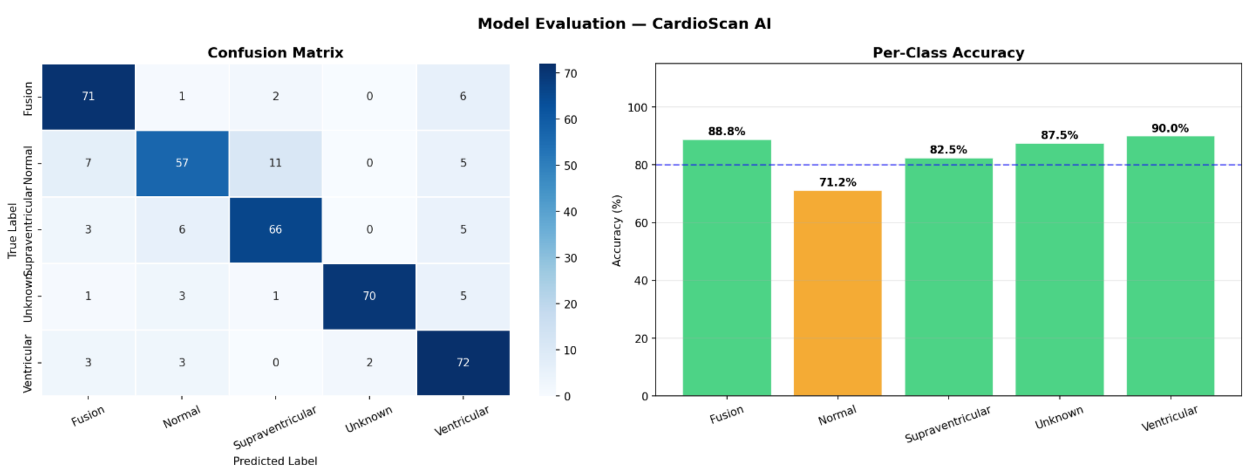
Insert table showing accuracy, precision, recall for each of the 5 classes (Normal, Supraventricular, Ventricular, Fusion, Unknown).

GPU Performance Analysis :

Training on H200 with BF16 mixed precision completed significantly faster than equivalent CPU-based training, demonstrating the value of GPU acceleration for medical imaging tasks.

Outputs :


CardioScan AI — Training Results
NVIDIA H200 GPU | ResNet-18 | ECG Classification
Training Loss
Train Loss
Loss
Epoch
Train vs Test Accuracy
Accuracy (%)
Epoch
Train Acc
Test Acc
Best: 84.0%
GPU Training Summary
Device: CUDA
Architecture: ResNet-18
Best Accuracy: 84.00%
Train Accuracy: 85.60%
Training Time: 1530.4s
Epochs: 15
Classes: 5
Batch Size: 32
Optimizer: AdamW
Loss Fn: CrossEntropy
Training Data Distribution
Number of Images
300
Fusion
Normal
Supraventricular
Unknown
Ventricular
Final Accuracy Comparison
Accuracy (%)
85.6%
83.2%
84.0%
Train Accuracy
Test Accuracy
Best Accuracy
Project Information
Team: HEARTNET
Project: CardioScan AI
Institution:
Presidency University
Program:
AI & GPU Summer Internship 2026
Hardware:
NVIDIA H200 Tensor Core GPU
Dataset: MIT-BIH ECG
Framework: PyTorch + CUDA


Model Evaluation — CardioScan AI
Confusion Matrix
True Label
Predicted Label
Fusion
Normal
Supraventricular
Unknown
Ventricular
71 1 2 0 6
7 57 11 0 5
3 6 66 0 5
1 3 1 70 5
3 3 0 2 72
Per-Class Accuracy
Accuracy (%)
88.8%
71.2%
82.5%
87.5%
90.0%

Conclusion & Future Work :

Conclusion :

This project successfully demonstrates that GPU-accelerated deep learning can effectively classify cardiac conditions from ECG images with high accuracy. The NVIDIA H200 GPU significantly reduced training time, enabling rapid experimentation with different model configurations.

Future Work :

- Expand the dataset with real hospital ECG records for better generalization

- Implement Explainable AI techniques like Grad-CAM to visualize model decision regions

- Integrate real-time ECG monitoring from wearable devices

- Explore ensemble models combining CNN and Transformer architectures

References :

1. MIT-BIH Arrhythmia Database, Kaggle

2. He, K. et al. "Deep Residual Learning for Image Recognition" (ResNet paper)

3. PyTorch Documentation

4. NVIDIA H200 Tensor Core GPU Datasheet